

1. (d)

Given, $\int_0^1 f(x) dx = 5$

Put $x = -t \Rightarrow dx = -dt$

$\therefore I = -\int_0^{-1} f(-t) dt = -\int_{-1}^0 f(t) dt \Rightarrow I = -5$

(3) $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, (where $a < c < b$)

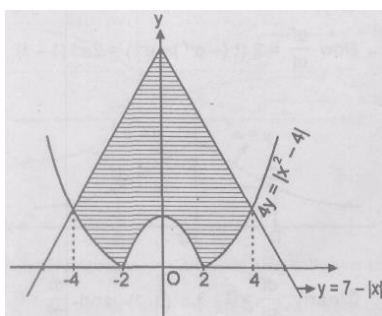
or $\int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \dots + \int_{c_n}^b f(x) dx$; (where $a < c_1 < c_2 < \dots < c_n < b$)

2. (d)

Required area,

$$= 2 \left(\int_0^2 \left(7-x - \left(\frac{4-x^2}{4} \right) \right) dx + \int_2^4 \left(7-x - \left(\frac{x^2-4}{4} \right) \right) dx \right)$$

$$= 2 \left(\int_0^2 \left(6-x - \frac{x^2}{4} \right) dx + \int_2^4 \left(8-x - \frac{x^2}{4} \right) dx \right)$$



$$= 2 \left(6x - \frac{x^2}{2} + \frac{x^3}{12} \right)_0^2 + 2 \left(8x - \frac{x^2}{2} - \frac{x^3}{12} \right)_2^4$$

$= 32$ sq. units.

3. (a)

Let $I = \int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x} = \int_{\pi/4}^{3\pi/4} \frac{dx}{1-\cos x}$

$\Rightarrow 2I = 2 \int_{\pi/4}^{3\pi/4} \frac{dx}{1-\cos^2 x} = 2 \int_{\pi/4}^{3\pi/4} \operatorname{cosec}^2 x dx = -2 \cot x \Big|_{\pi/4}^{3\pi/4} = 4$

$\Rightarrow I = 2$

4. (c)

We have,

$$\frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right) = \frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$\therefore \int_{-1}^1 \frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right) dx = \int_{-1}^1 \frac{-1}{1+x^2} dx = -2 \int_0^1 \frac{1}{1+x^2} dx$$

$$= -2 \left[\tan^{-1} x \right]_0^1 = -2 \left(\frac{\pi}{4} \right) = -\frac{\pi}{2}$$

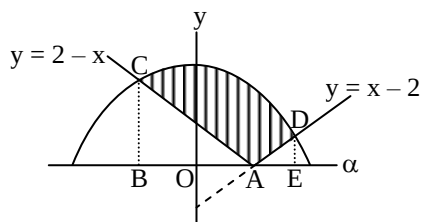
5. (b)

$3x^2 + 5y = 32$

$$\Rightarrow x^2 = -\frac{5}{3} \left(y - \frac{32}{5} \right) \quad y = |x - 2| = \begin{cases} x - 2, & \text{if } x > 2 \\ 2 - x, & \text{if } x < 2 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{x}{2} + \frac{y}{-2} = 1 \\ \frac{x}{2} + \frac{y}{2} = 1 \end{cases}$$

Here C (-2, 4), D(3, 1), A (2, 0)



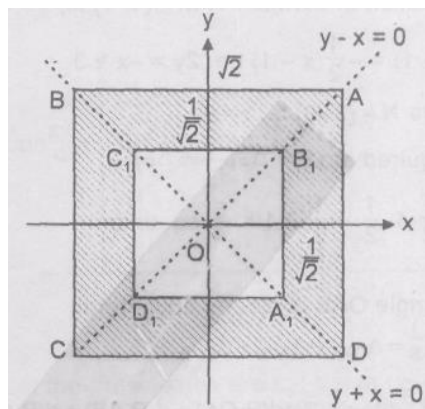
E(3, 0), B (-2, 0)

$$\text{Required area} = \int_{-2}^3 y dx - \Delta ABC - \Delta ADE$$

$$\begin{aligned} &= \int_{-2}^3 (2 - x^2) dx - \frac{1}{2} (BC \times AB) - \frac{1}{2} (DE \times AE) \\ &= \left[2x - \frac{x^3}{3} \right]_{-2}^3 - \frac{1}{2} (4 \times 4) - \frac{1}{2} (1 \times 1) \\ &= \left(6 - \frac{27}{3} \right) - \frac{1}{2} (16) - \frac{1}{2} (1) \\ &= (6 - 9) - 8 - \frac{1}{2} = -11 - \frac{1}{2} = -\frac{23}{2} \end{aligned}$$

6. (c)

For $y - x \leq 0$, $x + y \geq 0$.



We get, $\sqrt{2} \leq x + y + x - y \leq 2\sqrt{2}$

$$\Rightarrow \frac{1}{\sqrt{2}} \leq x \leq \sqrt{2}$$

for $y - x \geq 0$, $x + y \geq 0$ We get, $\frac{1}{\sqrt{2}} \leq y \leq \sqrt{2}$

For $x + y \leq 0$, $y - x \geq 0$ We get, $-\sqrt{2} \leq x \leq -\frac{1}{\sqrt{2}}$

For $x + y \leq 0$, $y - x \leq 0$ We get, $-\sqrt{2} \leq y \leq -\frac{1}{\sqrt{2}}$

Shaded region represents all positions of point P. its area is equal to $\Delta_{ABCD} - \Delta_{A_1B_1C_1D_1}$

i.e. $(2\sqrt{2})^2 - (\sqrt{2})^2$ i.e. 6 sq. units.

7. (a)

$$\int_{-1}^1 (x - [x]) dx = \int_{-1}^0 (x - [x]) dx + \int_0^1 (x - [x]) dx$$

$$\int_{-1}^0 (x+1)dx + \int_0^1 (x-0)dx$$

$$= \left[\frac{(x+1)^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} + \frac{1}{2} = 1.$$

8. (d)

$$\int_0^a f(x) g(x) h(x) dx$$

$$= \int_0^a f(a-x) g(a-x) h(a-x) dx$$

$$= - \int_0^a f(x) g(x) \left[\frac{3h(x)-5}{4} \right] dx$$

$$= - \frac{3}{4} \int_0^a f(x) g(x) h(x) dx + \frac{5}{4} \int_0^a f(x) g(x) dx$$

$$\frac{7}{4} \int_0^a f(x) g(x) h(x) dx = \frac{5}{4} \int_0^a f(x) g(x) dx = 0$$

$$\{f(a-x) g(a-x) = -f(x) g(x)\}$$

$$\text{So } \int_0^a f(x) g(x) h(x) dx = 0$$

9. (b)

$$\text{Put } e^x - 1 = t^2$$

$$\Rightarrow e^x dx = 2t dt$$

$$\Rightarrow I = \int_0^2 \frac{t \cdot 2t dt}{t^2 + 4}$$

10. (a)

$$f(x) = \sin^3 x |x| \Rightarrow f(-x) = \sin^3(-x) |-x|$$

$$= -\sin^3 x |x| \Rightarrow f(-x) = -f(x) \rightarrow \text{odd f}^n \therefore = 0$$

11. (b)

$$[\tan^{-1} x]_{-1}^1 =$$

$$\tan^{-1}(1) - \tan^{-1}(-1) = \frac{\pi}{4} - \left(-\frac{\pi}{4} \right) = \frac{\pi}{2}$$

12. (a)

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} dx = \sqrt{2} \int_0^{\pi/4} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \sqrt{2} \int_0^{\pi/4} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx$$

$$\text{Let } \sin x - \cos x = t$$

$$\Rightarrow (\cos x + \sin x) dx = dt$$

$$= \sqrt{2} \int_{-1}^0 \frac{dt}{\sqrt{1-t^2}} = \sqrt{2} (\sin^{-1} t)_{-1}^0$$

13. (d)

$\therefore$ function is odd function

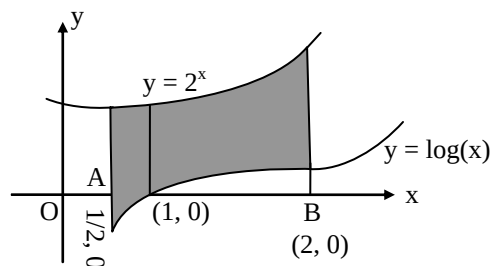
$$\int_{-a}^a f(x) dx = 0$$

14. (b)

Given curves are

$$y = \log x \dots (i)$$

$$y = 2^x \dots (ii)$$



So, the required area

$$= \left| \int_{1/2}^2 (y_1 - y_2) dx \right| = \left| \int_{1/2}^2 (2^x - \log x) dx \right|$$

$$= \left| \left[\frac{2^x}{\log 2} - (x \log x - x) \right]_{1/2}^2 \right|$$

$$= \left| \frac{4 - \sqrt{2}}{\log 2} - (2 \log 2 - 2) + \left(\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \right) \right|$$

$$= \frac{4 - \sqrt{2}}{\log 2} - \frac{5}{2} \log 2 + \frac{3}{2}.$$

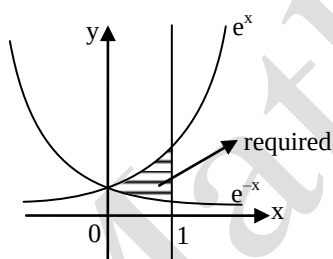
15. (b)

$$\text{Area} = \int_{-1}^1 \cos x \, dx = 2 \int_0^1 \cos x \, dx$$

$$= 2 \sin 1$$

16. (d)

$$A = \int_0^1 (e^x - e^{-x}) \, dx$$



$$\Rightarrow A = (e^x + e^{-x})_0^1$$

$$A = (e + e^{-1}) - (1 + 1)$$

$$A = e + \frac{1}{e} - 2$$

17. (a)

$$\text{Area} = \int_0^{\pi/4} \tan x \, dx + \int_0^{\pi/4} \cot x \, dx = \log \sqrt{2} + \log \sqrt{2} = \log 2$$

18. (b)

Using property P-4, we have

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

Adding it to given integral we have

$$2I = \int_0^{\pi/2} dx = [x]_0^{\pi/2} = \pi/2$$

$$\therefore I = \pi/4$$

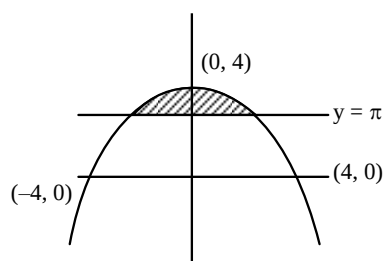
19. (c)

$$\lim_{x \rightarrow 0} \frac{1}{x} \left(\int_y^a e^{\sin^2 t} dt - \int_{x+y}^a e^{\sin^2 t} dt \right)$$

$$= \lim_{x \rightarrow 0} \frac{\int_y^{x+y} e^{\sin^2 t} dt}{x} = \lim_{x \rightarrow 0} \frac{e^{\sin^2(x+y)}}{1} = e^{\sin^2 y}$$

20. (c)

$$0 \leq \sin^2 x \leq 1$$



$$\Rightarrow -1 \leq -\sin^2 x \leq 0 \therefore [-\sin^2 x] = 0 \text{ or } -1 \text{ but } \sec^{-1}(0) \text{ is not defined hence } y = \sec^{-1}[-\sin^2 x] = \sec^{-1}(-1) = \text{now } \pi = \frac{16-x^2}{4} \Rightarrow x^2 =$$

$$16 - 4\pi = 4(4 - \pi) \Rightarrow x = \pm 2\sqrt{4 - \pi} \text{ The required area} = \int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}} \left(\frac{16-x^2}{4} - \pi \right) dx = \frac{8}{3} (4 - \pi)^{3/2}.$$